CS229 "Combat" Cheat Sheet

Machine Learning — Chapters 1–7 (PRML) — Rapid Exam Reference

1 Module 1: Probability Distributions

1.1 1.1 Univariate Gaussian

Formalism

PDF: $\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

$\mathbb{E}[x] = \mu, \quad \mathrm{Var}[x] = \sigma^2$

Precision form: $\beta = 1/\sigma^2$, so $\mathcal{N}(x|\mu,\beta^{-1}) \propto \exp\left(-\frac{\beta}{2}(x-\mu)^2\right)$

1.1.1 MLE (Computation)

Given $\{x_1,\ldots,x_N\}$:

$$\hat{\mu}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n \tag{1}$$

$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}(x_n-\hat{\mu})^2 \tag{2}$$

Log-likelihood:

$$\ln p(\mathcal{D}|\mu,\sigma^2) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2$$

MLE for σ^2 is **biased**: $\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{N-1}{N}\sigma^2$. Unbiased estimator divides by $(N-1)$, not N. For small N this matters significantly.

1.2 1.2 Multivariate Gaussian

Formalism

$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$

$\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}, \quad \mathrm{Cov}[\mathbf{x}] = \boldsymbol{\Sigma}$

Precision matrix: $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$

1.2.1 Sample Statistics

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n, \qquad \mathbf{S} = \frac{1}{N-1}\sum_{n=1}^{N}(\mathbf{x}_n-\bar{\mathbf{x}})(\mathbf{x}_n-\bar{\mathbf{x}})^T$$

PSD check: $\boldsymbol{\Sigma}$ is PSD iff all eigenvalues $\lambda_i \geq 0$. Solve $\det(\boldsymbol{\Sigma}-\lambda\mathbf{I}) = 0$.

High-D behavior: Most of the probability mass of a D-dimensional Gaussian lies in a thin *shell* at radius $\approx\sqrt{D}$ from the mean, not at the mode. The mode has *low* probability density in high D.

1.3 1.3 Beta-Bernoulli Conjugate Pair

Formalism

Likelihood: $p(x|\theta) = \theta^x(1-\theta)^{1-x}, \quad x \in \{0,1\}$

Prior: $p(\theta|\alpha,\beta) = \mathrm{Beta}(\theta|\alpha,\beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

Posterior (conjugacy): After m successes, ℓ failures:

$$p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta\,|\,\alpha+m,\ \beta+\ell)$$

Posterior mean: $\mathbb{E}[\theta|\mathcal{D}] = \frac{\alpha+m}{\alpha+\beta+m+\ell}$

Beta$(1,1)$ is the uniform prior on $[0,1]$. With this prior, the posterior mean $\frac{1+m}{2+m+\ell}$ acts as a Laplace-smoothed estimate. As $N\to\infty$, the posterior mean $\to$ MLE $= m/(m+\ell)$.

1.4 1.4 Categorical/Dirichlet

Prior: $\mathrm{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha})$. After counts m_k: $p(\boldsymbol{\mu}|\mathcal{D}) = \mathrm{Dir}(\boldsymbol{\mu}|\alpha_1+m_1,\ldots,\alpha_K+m_K)$.

2 Module 2: Linear Regression

2.1 2.1 Model & Design Matrix

Formalism

$y(\mathbf{x},\mathbf{w}) = \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}) = w_0 + \sum_{j=1}^{M-1} w_j\phi_j(\mathbf{x})$

Design matrix $(N \times M)$: $\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}$

For polynomial of degree $M-1$ in 1D: $\phi_j(x) = x^j$.

2.2 2.2 Ordinary Least Squares (OLS)

Minimize $E(\mathbf{w}) = \|\mathbf{t}-\boldsymbol{\Phi}\mathbf{w}\|^2$.

Normal equations (closed-form):

$$\boxed{\mathbf{w}_{\mathrm{ML}} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{t}}$$

$\boldsymbol{\Phi}^T\boldsymbol{\Phi}$ must be invertible. It is **singular** when: (a) $N < M$ (underdetermined), (b) columns of $\boldsymbol{\Phi}$ are linearly dependent, or (c) numerical ill-conditioning. Use regularization or pseudoinverse $\boldsymbol{\Phi}^+$.

2.3 2.3 Ridge Regression (Regularized)

Minimize $E(\mathbf{w}) = \|\mathbf{t}-\boldsymbol{\Phi}\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|^2$.

Ridge solution:

$$\boxed{\mathbf{w}_{\mathrm{ridge}} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi}+\lambda\mathbf{I})^{-1}\boldsymbol{\Phi}^T\mathbf{t}}$$

λ controls bias-variance trade-off:

- $\lambda\to 0$: OLS (low bias, high variance)
- $\lambda\to\infty$: $\mathbf{w}\to\mathbf{0}$ (high bias, low variance)
- $\lambda\mathbf{I}$ makes $(\boldsymbol{\Phi}^T\boldsymbol{\Phi}+\lambda\mathbf{I})$ always invertible

Over-regularization: predictions flatten toward the mean; check residuals on training data.

2.4 2.4 Bias-Variance Decomposition

For a test point $\mathbf{x}_*$ with true function $h(\mathbf{x}_*)$ and noise ϵ^2:

$$\boxed{\mathbb{E}[L] = \underbrace{(\bar{y} - h)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}_{\mathcal{D}}[y(\mathbf{x}_*)]}_{\text{variance}} + \epsilon^2}$$

where $\bar{y}(\mathbf{x}_*) = \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}_*)]$.
Computation: Given K datasets with predictions $y^{(k)}$: $\bar{y} = \frac{1}{K}\sum_k y^{(k)}$, $\mathrm{Var} = \frac{1}{K}\sum_k (y^{(k)} - \bar{y})^2$.

3 Module 3: Bayesian Linear Regression

3.1 3.1 Setup

Prior: $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$, Likelihood: $p(\mathbf{t}|\mathbf{w}) = \prod_n \mathcal{N}(t_n|\mathbf{w}^T\boldsymbol{\phi}_n, \beta^{-1})$.

3.2 3.2 Posterior

Core Formulas

$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N)$

$$\boxed{\mathbf{S}_N^{-1} = \alpha\mathbf{I} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}}$$

$$\boxed{\mathbf{m}_N = \beta\,\mathbf{S}_N\,\boldsymbol{\Phi}^T\mathbf{t}}$$

3.3 3.3 Predictive Distribution

At new point $\mathbf{x}_*$ with feature vector $\boldsymbol{\phi}_* = \boldsymbol{\phi}(\mathbf{x}_*)$:
$p(t_*|\mathbf{t}) = \mathcal{N}(t_*|\mu_*, \sigma_*^2)$

$$\boxed{\mu_* = \mathbf{m}_N^T\boldsymbol{\phi}_*, \quad \sigma_*^2 = \frac{1}{\beta} + \boldsymbol{\phi}_*^T\mathbf{S}_N\boldsymbol{\phi}_*}$$

$1/\beta$ = irreducible noise; $\boldsymbol{\phi}_*^T\mathbf{S}_N\boldsymbol{\phi}_*$ = epistemic (model) uncertainty. The latter shrinks with more data; the former does not.

3.4 3.4 Model Evidence (Marginal Likelihood)

Exact: $\ln p(\mathbf{t}|\alpha,\beta) = \frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - E(\mathbf{m}_N) - \frac{1}{2}\ln|\mathbf{S}_N^{-1}| - \frac{N}{2}\ln(2\pi)$
Simplified "box" approximation (per parameter):

$$\ln p(\mathcal{D}|\mathcal{M}) \approx \ln p(\mathcal{D}|\mathbf{w}_{\mathrm{MAP}}) + M\ln\frac{\Delta w_{\mathrm{post}}}{\Delta w_{\mathrm{prior}}}$$

The ratio $\Delta w_{\mathrm{post}}/\Delta w_{\mathrm{prior}} < 1$, so the second term is **negative** — it is the *Occam penalty*. More parameters $\Rightarrow$ bigger penalty. This naturally penalizes complex models.

3.5 3.5 Evidence Approximation

Eigenvalues λ_i of $\beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}$. Effective parameters:

$$\gamma = \sum_{i=1}^{M}\frac{\lambda_i}{\alpha + \lambda_i}$$

Update rules:

$$\alpha_{\mathrm{new}} = \frac{\gamma}{\mathbf{m}_N^T\mathbf{m}_N}, \qquad \beta_{\mathrm{new}}^{-1} = \frac{1}{N-\gamma}\sum_{n=1}^{N}(t_n - \mathbf{m}_N^T\boldsymbol{\phi}_n)^2$$

γ ranges from 0 to M. If $\gamma \ll M$, most parameters are "switched off" by the prior. If $\gamma \approx M$, data strongly constrains all parameters. Use λ_i of $\boldsymbol{\Phi}^T\boldsymbol{\Phi}$ (not $\beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}$) and multiply by β in the formula.

4 Module 4: Classification

4.1 4.1 Least-Squares Classification

Formalism

Same regression setup but $t_n \in \{0,1\}$: $y(\mathbf{x}) = \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x})$, classify as C_1 if $y(\mathbf{x}) \geq 0.5$.
Solution: $\mathbf{w} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{t}$ (same normal equations).

Problems with LS classification:
- Predictions not bounded to $[0,1]$
- Sensitive to outliers far from decision boundary
- Squared loss penalizes "too correct" predictions
- Can give worse boundaries than logistic regression

4.2 4.2 Logistic Regression

Formalism

$p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}))$ where $\sigma(a) = \frac{1}{1+e^{-a}}$
Properties: $\sigma(-a) = 1-\sigma(a)$, $\sigma'(a) = \sigma(a)(1-\sigma(a))$
Cross-entropy loss: $E(\mathbf{w}) = -\sum_{n=1}^{N}\left[t_n\ln y_n + (1-t_n)\ln(1-y_n)\right]$

4.2.1 Gradient & IRLS

$\nabla E(\mathbf{w}) = \sum_{n=1}^{N}(y_n - t_n)\boldsymbol{\phi}_n = \boldsymbol{\Phi}^T(\mathbf{y} - \mathbf{t})$
Hessian: $\mathbf{H} = \boldsymbol{\Phi}^T\mathbf{R}\boldsymbol{\Phi}$ where $\mathbf{R} = \mathrm{diag}(y_n(1-y_n))$.
IRLS update: $\mathbf{w}^{\mathrm{new}} = \mathbf{w}^{\mathrm{old}} - \mathbf{H}^{-1}\nabla E = (\boldsymbol{\Phi}^T\mathbf{R}\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{R}\mathbf{z}$
where $\mathbf{z} = \boldsymbol{\Phi}\mathbf{w}^{\mathrm{old}} - \mathbf{R}^{-1}(\mathbf{y} - \mathbf{t})$.

Vanishing gradients: When $y_n \approx 0$ or $y_n \approx 1$, $R_{nn} = y_n(1-y_n) \approx 0$, making $\mathbf{R}$ near-singular. This happens with perfectly separable data $\Rightarrow \|\mathbf{w}\| \to \infty$. Fix: add regularization $E(\mathbf{w}) + \frac{\lambda}{2}\|\mathbf{w}\|^2$.

Gradient descent: $\mathbf{w}^{\mathrm{new}} = \mathbf{w}^{\mathrm{old}} - \eta\nabla E$. For logistic regression the loss is convex, so any η small enough converges. Typical: $\eta \in [0.001, 0.1]$. Decrease η if loss oscillates. IRLS (Newton's method) converges quadratically but each step is $O(M^3)$.

5 Module 5: Special-Case & Edge-Case Matrices

5.1 5.1 Regression Edge Cases

Scenario	Symptom & Fix
$N < M$	$\mathbf{\Phi}^T\mathbf{\Phi}$ singular. Use ridge ($+\lambda\mathbf{I}$) or reduce features.
Collinear cols	$\det(\mathbf{\Phi}^T\mathbf{\Phi}) \approx 0$. Weights blow up. Regularize or PCA.
λ too large	Underfitting: all $w_j \to 0$. Residuals large, flat prediction.
λ too small	Overfitting: weights large, low train error, high test error.

5.2 5.2 Classification Edge Cases

Scenario	Symptom & Fix
Perfect sep.	$\|\mathbf{w}\| \to \infty$ in logistic reg. Add L_2 penalty.
LS outlier	LS classification boundary shifts. Logistic is robust.
Class imbal.	Prior π_k matters. Adjust threshold or re-weight loss.
High-D	Curse of dimensionality. Most volume near surface of hypercube.

5.3 5.3 Gaussian Edge Cases

Scenario	Symptom & Fix
Singular $\mathbf{\Sigma}$	Degenerate Gaussian. Add small $\epsilon\mathbf{I}$ or reduce D.
$N \leq D$	Sample covariance is rank-deficient. Use shrinkage estimator.
High D	Mahalanobis dist. concentrates near $\sqrt{D}$. Mode $\neq$ typical set.
MLE bias	$\hat{\sigma}^2_{\text{ML}}$ underestimates. Use $N-1$ correction for small N.

6 Module 6: Quick-Reference Formulas

6.1 6.1 Matrix Calculus Identities

$$\frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^T\mathbf{a}) = \mathbf{a} \tag{3}$$

$$\frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^T\mathbf{A}\mathbf{w}) = (\mathbf{A} + \mathbf{A}^T)\mathbf{w} \tag{4}$$

$$\frac{\partial}{\partial \mathbf{w}}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 = -2\mathbf{\Phi}^T(\mathbf{t} - \mathbf{\Phi}\mathbf{w}) \tag{5}$$

If $\mathbf{A}$ symmetric: $\frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^T\mathbf{A}\mathbf{w}) = 2\mathbf{A}\mathbf{w}$.

6.2 6.2 Woodbury & Useful Identities

$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}$$

$$|\mathbf{A} + \mathbf{u}\mathbf{v}^T| = |\mathbf{A}|(1 + \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u})$$

6.3 6.3 Eigenvalue Quick Check for 2×2

For $\mathbf{A} = ab$
cd: $\lambda = \frac{(a+d)\pm\sqrt{(a+d)^2-4(ad-bc)}}{2}$
PSD iff $a \geq 0$, $d \geq 0$, and $ad - bc \geq 0$.

6.4 6.4 Convergence Criteria Checklist

1. $|E^{(t)} - E^{(t-1)}|/|E^{(t)}| < \epsilon$ (relative change)
2. $\|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\| < \epsilon$ (parameter change)
3. $\|\nabla E\| < \epsilon$ (gradient norm)
4. For EM/evidence: monitor log marginal likelihood

"All models are wrong, but some are useful." — G.E.P. Box